MILITARY SPECIFICATION

METERING SYSTEM, STORAGE BATTERY CURRENT AND AMPERE-HOUR, DIGITAL (METRIC)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers a digital current and ampere-hour metering system for a storage battery circuit.

1.2 Classification. Meters shall be of the classes specified in 1.2.1 and of the types specified in 1.2.2.

1.2.1 Class. Classes shall be determined by the application (see 3.3); and shall be as follows:

Class I - Submarine application - Without ampere display.
Class II - Surface ship application - With ampere display.

1.2.2 Type. Type shall be determined by the shunt rating values (see 3.3). Types shall be as follows:

Type I - 2,500 amp/100 mV shunt
Type II - 3,000 amp/100 mV shunt
Type III - 4,000 amp/100 mV shunt
Type IV - 5,000 amp/100 mV shunt
Type V - 10,000 amp/100 mV shunt

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

L-P-393 - Plastic Molding Material, Polycarbonate, Injection and Extrusion.

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact), Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-915 - Cable and Cord, Electrical, For Shipboard Use, General Specification for.
MIL-C-915/49 - Cable, Electrical Type 2SWU (Including Variation 2SWUA).
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-E-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-E-16400 - Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore: General Specification for.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.
MIL-S-19622 - Stuffing Tubes, Nylon, General Specification.
MIL-C-21097 - Connectors, Electrical, Printed Wiring Board, General Purpose, General Specification for.
MIL-J-24142 - Junction Boxes for Electrical Fittings and Fixtures, General Specification for.
MIL-E-24142/3 - Enclosure for Electrical Fittings and Fixtures, Submersible, Size 6 By 9.
MIL-I-45208 - Inspection System Requirements.
MIL-T-55164 - Terminal Boards, Molded, Barrier, Screw and Stud Types, and Associated Accessories, General Specification for.

STANDARDS

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-470 - Maintainability Program Requirements (For Systems and Equipments).
MIL-STD-781 - Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.

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MILITARY (Continued)

MIL-STD-1364 - Standard General Purpose Electronic Test Equipment.

DOD-STD-1399, Section 300 - Interface Standard for Shipboard Systems, Electric Power, Alternating Current (Metric).

2.1.2 Government publications. The following Government publication forms a part of this specification to the extent specified herein.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

0967-LP-00-312-8010 - Maintainability Design Criteria Handbook for Designers.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

Y32.16 - Electrical and Electronic Parts and Equipment Reference Designation for. (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

2.3 Order of precedence. In the event of a confict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 System assemblies. The current measuring system shall consist of the following assemblies:

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(a) Ammeter. The ammeter shall be a digital instrument of the specified class (see 1.2.1) and type (see 1.2.2), capable of sensing the voltage across the appropriate shunt and connecting this signal into digital information for use in providing a direct reading display of amperes when required (see 1.2.1) and for providing binary - coded - decimal electrical outputs for the purpose of integration by the ampere-hour meters described in 3.3(b). When a class II meter is required (see 1.2.1) it shall consist of a polarity sign and at least four full decimal digits. The range of the ammeter shall permit the display of currents at least as large as ± 9,999 amperes with a resolution of one ampere, and a display of currents at least as large as ± 999.9 amperes with a resolution of one-tenth ampere. This may be accomplished with a 4-digit or a 4-1/2-digit display auto-ranging, or with a 5-digit display and a fixed decimal point to the left of the least significant digit. If a 5-digit display is not provided, the range displayed shall be indicated to the operator by the presence or absence of the decimal point to the left of the least significant digit. Except when currents of less than one ampere are displayed using one zero before the decimal point, display of leading zeros, shall be suppressed. Indication of overranging on currents greater than the maximum value which the meter is capable of displaying shall be provided. Indication of overranging shall also be provided when a display of amperes is not required (see 1.2.1).

(b) Ampere-hour meters. Two, non-mechanical, digital, ampere-hour meters shall be provided. One of these meters shall digitally display the integral of the positive-polarity output and the other shall display the integral of the negative-polarity output of the ammeter of 3.3(a). Except for the front-panel legends, "charge" or "discharge", the two ampere-hour meters shall be identical and interchangeable. The meters shall display the ammeter output to a maximum value of 999,999.9 ampere-hours. A pushbutton switch shall be provided on the front panel of each meter which will permit the operator to clear the accumulated value to zero. The outputs of the ampere-hour meters shall be in the form of a seven-digit, decimal display with the decimal point to the left of the least significant digit. The display shall suppress leading zeros, and an indication of overflow beyond 999,999.9 ampere-hours shall be provided. Each ampere-hour meter shall also provide an electrical output for the purpose of operating the totalizing indicators described in 3.3(c). The ampere-hour meters shall incorporate CMOS latches and a capacitor or an internal standby battery, charged from the alternating current (a.c.) power line and capable of retaining the accumulated ampere-hour reading for a period of at least one week in the event of a power line failure. The standby battery if used need not operate the display or the integration circuits; however, there shall be no loss of stored data upon power interruption or restoration.

(c) Totalizing ampere-hour indicators. Two totalizing ampere-hour indicators shall be provided. These shall be digital instruments, one indicating charge ampere-hours and the other

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indicating discharge ampere-hours. The totalizing indicators shall consist of non-mechanical counters designed to accept electrical input pulses from the ampere-hour meters of 3.3(b). Each pulse shall correspond to a charge or discharge of one-tenth ampere-hour and the totalizing indicator shall accumulate these pulses and display the result in the form of seven decimal digits to a maximum value of 9,999,999 ampere-hours. Totalizing ampere-hour displays shall be concealed from the operators normal field of view by a spring loaded cover. Except for the front-panel legends, "charge" or "discharge", the two totalizing indicators shall be identical and interchangeable. Leading zeroes, except for integer ampere-hours, shall be suppressed in the display, and an indication of overflow beyond 9,999,999 ampere-hours shall be provided. The totalizing indicator shall be capable of being manually cleared to zero by means of a pushbutton switch, accessible by removing the front face of the indicator. Each totalizing indicator shall incorporate CMOS latches and a capacitor or a standby battery for data retention in the event of power interruption as described in 3.3(b).

3.4 Connecting cable. The connections between the current shunt, the ammeter, and the totalizing indicators will be made by shielded cables having a maximum resistance of 2.3 ohms per conductor per 100 meters at 25 degrees Celsius (°C) (type 2SWU of MIL-C-915 and MIL-C-915/49). Unless otherwise specified (see 6.2.1), cables shall not be furnished. Maximum cable lengths will be as follows:

(a) Between the current shunt and the ammeter - 100 meters.

(b) Between the ampere-hour meter and totalizing ampere-hour indicator - 30 millimeters (mm).

3.4.1 Stuffing tubes. Stuffing tubes shall conform to MIL-S-19622.

3.4.2 Circuit board pin assignments. Connector pins shall be assigned on the ammeter, ampere-hour meter, and totalizing ampere-hour indicators as shown in tables I, II, and III, respectively.

TABLE I. Ammeter connector pin designations.

Ammeter A to D converter board connector	
15-position edgeboard connector	
PIN FUNCTION	PIN FUNCTION
1 NC	A NC
2 ANALOG INPUT	B NC
3 NC	C NC
4 NC	D NC
5 NC	E NC

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TABLE I. Ammeter connector pin designations. - Continued

Ammeter A to D converter board connector	
15-position edgeboard connector	
PIN FUNCTION	PIN FUNCTION
6 NC	F NC
7 NC	H NC
8 NC	J NC
9 NC	K NC
10 ANALOG COMMON	L NC
11 NIGH RANGE	M NC
12 NC	N DIGITAL GND
13 NC	P NC
14 NC	R EARTH GND
15 AC POWER	S AC POWER RETURN
Ammeter digital board connector	
10-position edgeboard connector	
PIN FUNCTION	PIN FUNCTION
1 BCD 4	A BCD 1
2 DIGITAL GND	B BCD 2
3 BCD 8	C BCD 10
4 BCD 40	D BCD 20
5 BCD 80	E BCD 100
6 BCD 400	F DATA VALID
7 BCD 800	H BCD 200
8 BCD 4,000	J BCD 1,000
9 BCD 8,000	K BCD 2,000
10 POLARITY	L BCD 10,000

TABLE II. Ampere-hour meter connector pin designations.

Ampere-hour meter connector	
15-position edgeboard connector	
PIN FUNCTION	PIN FUNCTION
1 BCD 4	A BCD 1
2 DIGITAL GND	B BCD 2
3 BCD 8	C BCD 10
4 BCD 40	D BCD 20
5 BCD 80	E BCD 100
6 BCD 400	F LOAD
7 BCD 800	H BCD 200

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TABLE II. Ampere-hour meter connector pin designations. - Continued

Ampere-hour meter connector			
15-position edgeboard connector			
PIN	FUNCTION	PIN	FUNCTION
8	BCD 4,000	J	BCD 1,000
9	BCD 8,000	K	BCD 2,000
10	HI-RANGE SELECT	L	BCD 10,000
11	HOLD	M	NC
12	NC	N	HOLD
13	OUTPUT (HIGH)	P	OUTPUT (COMMON)
14	NC	R	NC
15	AC POWER	S	AC POWER RETURN

TABLE III. Totalizing ampere-hour indicator connector pin designations.

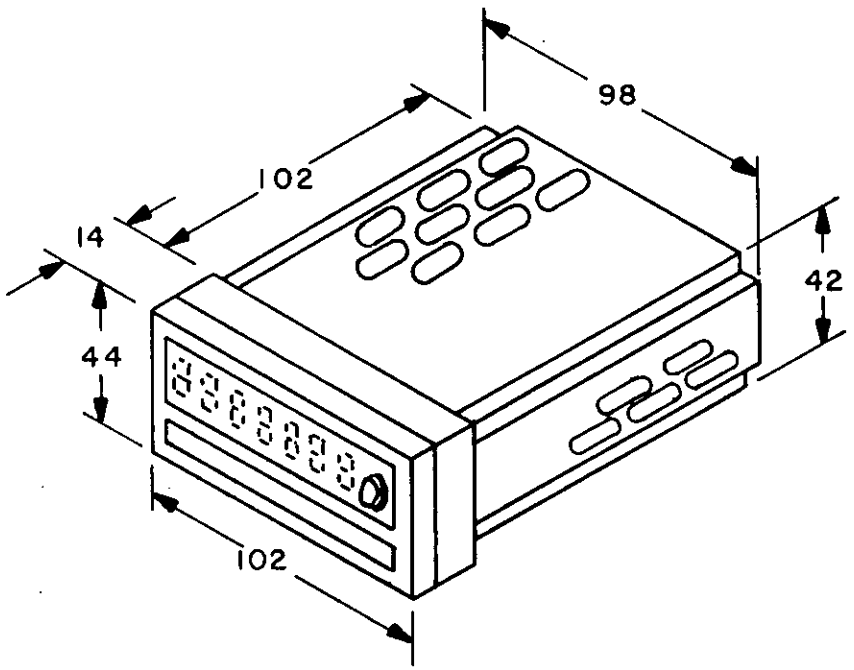
Totalizing ampere-hour indicator connector			
15-position edgeboard connector			
PIN	FUNCTION	PIN	FUNCTION
1	NC	A	NC
2	NC	B	NC
3	NC	C	NC
4	NC	D	NC
5	NC	E	NC
6	NC	F	NC
7	NC	H	NC
8	NC	J	NC
9	NC	K	NC
10	NC	L	NC
11	NC	M	NC
12	NC	N	NC
13	INPUT (HIGH)	P	INPUT (COMMON)
14	NC	R	NC
15	AC POWER	S	AC POWER RETURN

3.5 Construction of assemblies.

3.5.1 Instruments. The ammeter, ampere-hour meters, and totalizing ampere-hour indicators shall all be constructed in identical, digital-panel-meter cases (see figure 1). The cases shall be molded from a plastic material in accordance with L-P-393. Printed wiring boards within the instruments shall be in accordance with requirement 17 of MIL-STD-454. The display digits shall be of the luminous type [either plasma discharge or light emitting diode (LED)], orange in color, at least 1 centimeter (cm) in height, and easily readable in normal room illumination over an angular range of at least plus or minus 45 degrees from directly ahead. Batteries and other components of

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limited life shall be socketed for ease of replacement and secured against becoming dislodged by shock or vibration. Circuit board contact fingers used for interconnection with edgeboard connectors shall be gold plated. The construction shall include, to the greatest extent possible, features that facilitate maintenance. The front panel of the ammeter shall carry the legend: "AMPERES." The front panels of the ampere-hour meters and the totalizing ampere-hour indicators shall carry the legend: "CHARGE AMPERE-HOURS" or the legend: "DISCHARGE AMPERE-HOURS", as required. The pushbutton on the front panel of the ampere-hour meter shall be labeled with the legend: "RESET". Provision shall be made for securing the instruments to the cover of the junction box by means of screws, accessible from the front of the instrument by removing the front faces of the instruments.



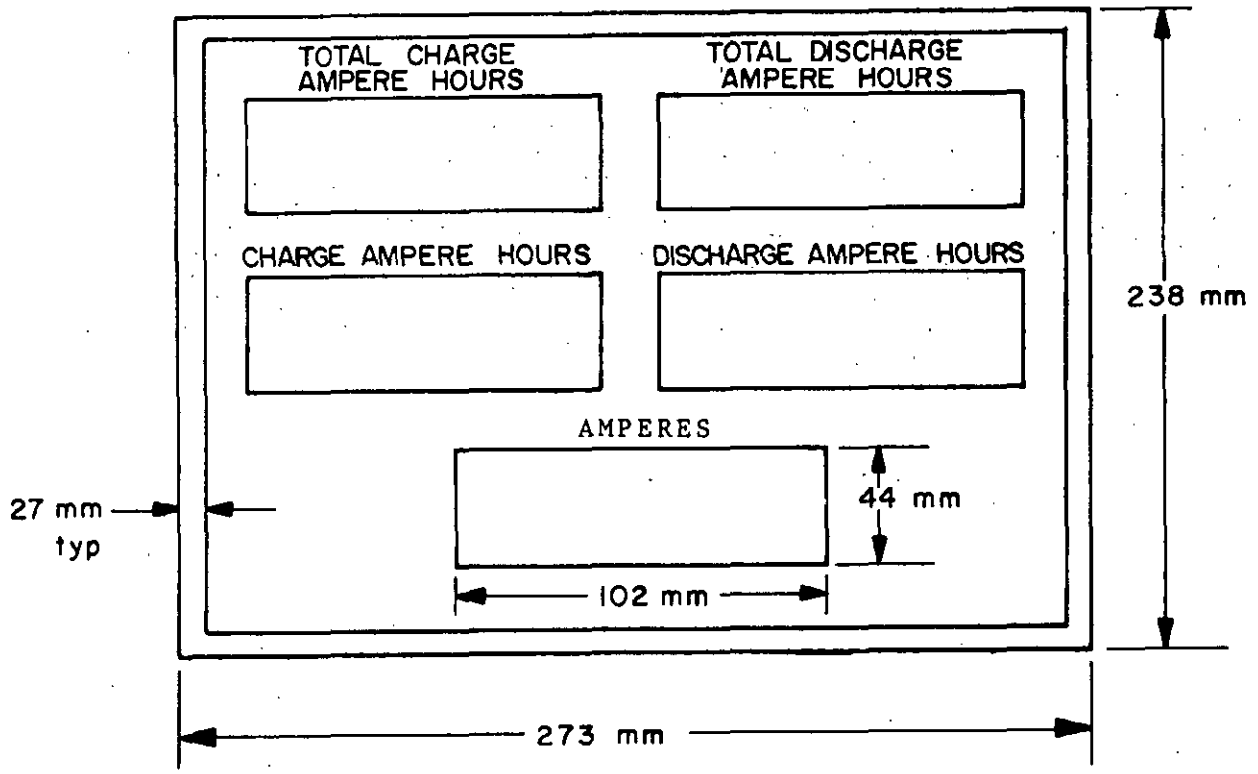
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NOTES:
1. Dimensions are in mm.
2. Unless otherwise specified, tolerances are plus or minus 1 mm.

FIGURE 1. Outline dimensions.

3.5.2 Meter mounting. The ammeter and the two ampere-hour meters shall be installed in a junction box in accordance with MIL-J-24142 and MIL-E-24142/3, and two totalizing ampere-hour indicator meter arrangements shall be in accordance with figure 2. A prewired mounting plate containing the edgeboard connectors which mate with the circuit boards protruding from the rear of the instruments shall be installed in each of the junction boxes and shall support the rear of the instruments.

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NOTES:
1. Dimensions are in mm.
2. Depth dimension is 229 mm.
3. Unless otherwise specified, tolerances are plus or minus 1 mm.

FIGURE 2. Instrument arrangement.

3.5.3 Edgeboard connectors. The edgeboard connectors shall be in accordance with MIL-C-21097.

3.6 System connections and isolation. The junction box shall be provided with stuffing tubes and terminal blocks as required to make electrical connections to the battery bus, the alternating current (a.c.), power source, and the instrument mounting panels. The battery bus current shunt shall be connected to the terminal board within the junction box which is normally inaccessible to the operators; but may be exposed while these instruments are being replaced or adjusted for calibration purposes. A minimum insulation of 10 megohms and a minimum dielectric withstand voltage of 1,500 volts root mean square (V rms) 60 hertz (Hz) shall be maintained between the battery bus, the a.c. power line, and the instrument mounting panels. There shall be no evidence of arcing, corona (visible, audible, or odorous), flashover, or punctured insulation following the dielectric withstand test.

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3.7 System performance.

3.7.1 Accuracy. The accuracy of the current measuring system shall be:

(a) For current (class II only) - plus or minus 0.3 percent of reading.
(b) For ampere-hours - plus or minus 0.3 percent of reading multiplied by time in hours.

These accuracies shall apply under the following reference conditions:

(a) Current - zero to 10,000 amperes.
(b) Temperature - 20 to 30°C.
(c) Relative humidity - 10 to 60 percent.
(d) Power line voltage - 110 to 120 V a.c.
(e) Power line frequency - 57 to 63 Hz.
(f) Operating position - any.
(g) Warm-up time - not more than 5 minutes.

For errors due to environmental conditions (see 3.10).

3.7.2 Input power. Instruments of the current measuring system shall be connected to and capable of operating from electrical power having characteristics as specified for type I of DOD-STD-1399, section 300. The display indications shall not be affected to an extent greater than that specified in 3.10. System calibration shall not be permanently affected by voltage or frequency excursions specified in 3.10. The power required from the 60 Hz source to operate the system shall not be more than 50 watts (W) under the reference conditions of 3.7.1.

3.7.3 Busbar current. Although over-range point for the current measuring system shall be at least 10,000 amperes, no damage to the system instruments or permanent effect upon system calibration shall result from fault currents to 20,000 amperes of any duration or from transient changes in busbar currents at any rate. This tolerance does not necessarily apply to the current measuring shunt, which is not a supplied part of the system.

3.7.4 Busbar voltage. The nominal busbar voltage is 270 volts direct current (V d.c.), with a range of 175 V d.c. to 355 V d.c. with one battery of 126 or 128 cells.

3.7.5 Calibration. All calibration adjustments for the system shall be located within the ammeter case. Calibration shall be unaffected by variations in the lengths or resistance of cables between the system instruments or between the system and the current-measuring shunt so long as these values remain within the limits of 3.4. Calibration adjustments shall be unaffected by shock, vibration, or other environmental conditions within the limits specified herein.

3.8 Environmental requirements.

3.8.1 Mechanical.

(a) Shock. System assemblies, subassemblies, and parts shall meet the requirements for type A, grade A, hull (bulkhead) mounted,

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lightweight, class I equipment as specified in MIL-S-901. Change in indication of the meters and the indicator (following reset of the meter to zero) shall not exceed the value specified in 3.10. Following the shock test of 4.7.2.4, the system shall show no signs of breakage or distortion of parts sufficient to impair operation. Insulation resistance shall be not lower than the value of 3.6 and dielectric withstand voltage shall be not lower than 65 percent of the values of 3.6.

(b) Vibration. System parts shall withstand the type I vibration requirements of MIL-STD-167-1 without change in indication of the instruments and the indicator error (following reset of the meter to zero) shall not exceed the value specified in 3.10. Following the vibration test of 4.7.2.5 the system shall show no signs of breakage or distortion of parts sufficient to impair operation. Insulation resistance shall be not lower than the values in 3.6.

3.8.2. Temperature range. The system shall be capable of operating in the temperature range of 0 to 50°C. Reference temperature of initial calibration shall be 26.7 ± 2°C. Changes in display indications due to ambient temperature excursions within 0 and 50°C shall not exceed the values specified in 3.10. The system shall be capable of storage at temperatures in the range minus 20 to plus 70°C without damage.

3.8.3 Relative humidity. The ammeter, ampere-hour meter, and totalizing indicator shall be capable of storage for 168 hours at 95 percent plus or minus 5 percent relative humidity. Following such storage, insulation resistance shall be not less than 2 megohms and the components shall show no signs of material damage or corrosion.

3.8.4 Watertightness. The system shall meet the watertight requirements of MIL-E-2036.

3.9 Inclination. The system operation, performance, and accuracy shall not be affected by mounting position or inclination and shall be capable of meeting the requirements of MIL-E-16400.

3.10 Errors due to environmental conditions. In addition to the errors permitted in 3.7.1, errors as shown in table IV also shall be permitted under or immediately following exposure to the extremes of temperature or line power. Requirements for the ammeter apply to class II instruments only (see 1.2.1).

TABLE IV. Errors due to environmental conditions.

Condition	Instrument	Reading
Temperature 0 to 20°C 30 to 50°C	Ammeter Ampere-hour meter and totalizing ampere-hour indicator	± 0.5 percent of reading ± 0.5 percent of reading Multiplied by time in hours

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TABLE IV. Errors due to environmental conditions. - Continued

Condition	Instrument	Reading
Input power line voltage or frequency variation greater than 3 percent	Ammeter Ampere-hour meter and totalizing ampere-hour indicator	± 0.5 percent of reading ± 0.5 percent of reading Multiplied by time in hours
Vibration	Ammeter Ampere-hour meter and totalizing ampere-hour indicator	± 0.5 percent of reading ± 0.5 percent of reading Multiplied by time in hours
Shock	Ammeter Ampere-hour meter and totalizing ampere-hour indicator	± 0.5 percent of reading ± 0.5 percent of reading Multiplied by time in hours

3.11 Technical data. The contractor shall prepare manuals and drawings in accordance with the data ordering documents (see 6.2.2).

3.11.1 Drawings. In addition to the drawing requirements covered by the data ordering document (see 6.2.2) the unique technical features specified in 3.11.1.1 shall be included.

3.11.1.1 Drawings and certification data. Drawings shall not be smaller than 430 by 560 mm nor larger than 710 by 1020 mm. The size drawing scale shall be such that the number of sheets required shall be the minimum consistent with content and clarity. Drawings shall contain the following data:

(a) Manufacturer's name and model number.
(b) Reference to this specification.
(c) List of parts, tabulating part identification, including original manufacturer's part number, specification data of part, find number corresponding to part flags of outline drawings.
(d) Identification and stress level data (part manufacturer's and in-use ratings) of each electrical part.
(e) Schematic diagrams.
(f) Shock classification and approval reference.
(g) Outline drawing of each assembly including:
    (1) Overall and mounting dimensions
    (2) Location and size of mounting holes
    (3) Mass of each assembly

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(4) Panel cutout data for the indicator
(5) Position, size and marking of terminals used for power input and system part interconnection.

(h) Interconnection cable requirements such as permissible resistance, and shielding requirements.
(i) Amperes and power required from the 115 V, 60 Hz source.
(j) Graphic or tabular presentation of characteristics of circuits, meters, and indicator. This data is required for trouble-shooting assistance and contain complete calibration procedures and instructions. Normal readings at relevant circuit points under specified condition shall be presented.
(k) List of recommended test equipment such as multimeters or portable standard instruments.
(l) Schematic circuit of the complete system.
(m) Connection diagram showing approximate part location, cable routing within equipment and terminal and wire marking.
(n) Description of operation, describing briefly how the circuit parts perform their functions to yield specified systems operation.

3.12 Maintainability. The equipment shall be maintained by easy replacement of removable parts or modules. The design shall conform to the design-for-maintainability requirements of MIL-STD-470 and publication NAVSEA 0967-LP-00-312-8010. Positioning and alinement of all instruments shall employ positive means, so that correct reassembly is repeatedly assured. In no case shall connectors for a given instrument be physically interchangeable unless the instrument is also interchangeable with regard to function and performance. The equipment repair time (ERT) shall be not more than 1 hour.

3.12.1 Test equipment. The system shall be capable of being maintained, repaired, and calibrated with the aid of the following test equipment selected from MIL-STD-1364.

(a) Multimeter.
(b) Oscilloscope.
(c) D.c. voltage calibrator.
(d) Clock or watch.

3.12.1.1 Unspecified test equipment. A contractor desiring to use test equipment other than that specified herein shall submit the list to NAVSEA with justification of the necessity for using such test equipment prior to the start of manufacture of the system.

3.12.2 Repair of removable parts and subassemblies. The repairability of any removable modules, parts or subassemblies shall be determined in accordance with MIL-E-917. The time required to repair any malfunctioned parts, module or subassembly, after removed from the system, shall be not more than 10 minutes.

3.13 Reliability. The current measuring system shall have a mean-time-between-failure (MTBF) of not less than 10,000 hours. Definition of failure shall be increase of errors of indication beyond the limits shown in 3.7.1 or failure to indicate.

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3.14 Identification of equipment division by reference designation. To provide identification that will aid documentation and to facilitate maintenance, each part, subassembly or assembly of the current measuring system shall be identified by reference designations in accordance with ANSI Y32.16.

3.15 Identification plate. The current measuring system shall be equipped with an identification plate in accordance with MIL-P-15024. The identification plates shall conform to the marking requirements of MIL-P-15024/5.

3.16 Safety. Safety shall be in accordance with MIL-STD-454, requirement 1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination and tests specified in table V.

TABLE V. Inspections.

Inspection	First article inspection	Quality conformance inspection	Requirement paragraph	Inspection paragraph
General examination	X	X	3.5	4.7.1
Initial accuracy	X	X	3.7.1	4.7.2.1
Input power variation	X	X	3.7.2	4.7.2.2
Sustained high current and short circuit	X		3.7.3	4.7.2.3
Shock	X		3.8.1(a)	4.7.2.4

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TABLE V. Inspections. - Continued

Inspection	First article inspection	Quality conformance inspection	Requirement paragraph	Inspection paragraph
Vibration	X		3.8.1(b)	4.7.2.5
Inclination	X		3.9	4.7.2.7
Temperature cycling	X		3.8.2	4.7.2.6
Insulation resistance	X	X	3.6	4.7.3
Humidity	X		3.8.3	4.7.3.1
Dielectric withstand voltage	X	X	3.6	4.7.4
Calibration	X	X	3.7.5	4.7.6
Maintainability demonstration	X		3.12	4.8
Enclosure	X		3.5.1 and 3.5.2	4.10
Reliability		X	3.13	4.9

4.3.1 First article inspection report. A first article inspection report shall be prepared in accordance with the data ordering documents as specified (see 6.2.2).

4.4 Quality conformance inspection. Each current measuring system offered for delivery shall be subjected to the quality conformance inspection as specified in table V.

4.5 Test conditions. Unless otherwise specified in the applicable test paragraph, all tests shall be conducted under the conditions specified in 3.7.1, with the exception of temperature cycling test (see 4.7.2.6) which may be conducted at ambient temperature of 25 ± 2°C and corrections to 26.7 ± 2°C (reference conditions shall be made to all readings and measurements). The specific ambient temperature at the time of test shall be measured and recorded. Necessary adjustment and calibration of the system shall be made prior to starting of the tests, no adjustments or calibration shall be allowed during or between the tests.

4.6 Test plan and test report. The contractor shall prepare a test plan and a test report in accordance with the data ordering documents (see 6.2.2).

4.7 Inspection procedure.

4.7.1 Visual examination. Units and assemblies of the current measuring system shall be visually examined to determine that:

(a) Parts used are as identified by drawings and fabrication techniques.
(b) Mounting of parts is as shown on drawings. This includes examination for correct location, orientation, securing method (fasteners, insulators, and others) and, where applicable, torquing.

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(c) Marking of parts (or, where parts are too small to mark, marking of adjacent panel) and terminals is complete and consistent with schematic and connection diagrams.
(d) Hookup wire is as shown on drawings.
(e) Hookup wire is neatly formed into groups and laced, tied or clamped securely into place (with nonflammable cord or clamps) to prevent chafing due to vibration.
(f) Wire ends are marked.
(g) Stuffing tubes are provided where wires pass through holes in metal panels.
(h) Wires are not bent sharply around corners.
(i) Wires are connected by solder or other mechanical means (with locking devices) connections.
(j) Indicator mounting panel drilling plan and mounting means as specified.
(k) Soldering joints are properly done.

4.7.2 Accuracy determination.

4.7.2.1 Initial. The current measuring system shall be operated at conditions specified in 3.7.1 and readings made at busbar current and ampere-hour values of table VI.

TABLE VI. Initial accuracy.

Charge busbar current (amperes)	Discharge busbar current (amperes)	Ampere-hours (minimum)
1	1	3
5 1/	5 1/	2.5 1/
50 1/	50 1/	5 1/
250	250	125
500 1/	500 1/	250
800	800	250
1,000	1,000	50 1/
2,500	2,500	250
5,000 1/	5,000 1/	100 1/
6,000 1/	8,000 1/	100 1/

1/ Units after first article tested to these values only.

4.7.2.2 Input power variations. Input voltage and frequency shall be adjusted to the following values to check the system accuracy at 5, 500, and 800 charge amperes and 800 discharge amperes busbar current (use table VI for the minimum ampere-hours required for each value of current):

(a) 103.5 V at 63 Hz.
(b) 103.5 V at 57 Hz.
(c) 126.5 V at 63 Hz.
(d) 126.5 V at 57 Hz.

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4.7.2.2.1 If data shows that errors are equal to or less than 2/3 of the sum of the applicable values of 3.7.1 or table IV, no additional values of busbar current need be checked. Otherwise, each value of table IV shall be checked. Voltage shall also be adjusted to 103.4 V at 60 Hz, then 126.5 V at 60 Hz for 15 minutes each, then indication at 50 amperes of busbar current checked.

4.7.2.3 Sustained high current fault current. The accuracy test of 4.7.2.2 shall be made after the system has been operated at currents (of equivalent ampere-turn) for the periods as follows:

(a) 8,000 amperes for 10 minutes.
(b) 20,000 amperes for 1 second.

4.7.2.4 Shock. The ammeter and ampere-hour indicator installed in junction boxes shall be mounted on a fixture fabricated to simulate an actual installation. The ampere-hour indicator shall be mounted on a portion of the fixture similar to fixture 6D-1 of MIL-S-901. The assembly shall then be tested as specified in MIL-S-901. Tests shall be performed with approximately 50 amperes of busbar current flowing.

4.7.2.4.1 Definition of failure of shock test. Following the shock test of 4.7.2.4, failures shall be defined as follows:

(a) Failure to indicate or errors of indication exceeding the values of table IV as determined by the accuracy checks of 4.7.2.2.
(b) Breakage or distortion of any parts sufficient to impair operation.
(c) Insulation resistance lower than the value of 3.6.
(d) Dielectric withstand voltage lower than 65 percent of the specified value of 3.6.

4.7.2.5 Vibration. Using the same fixture as specified in 4.7.2.4, the system assemblies and subassemblies shall be subjected to a type I test of MIL-STD-167-1. Failure of this test shall be determined by the same criteria as in 4.7.2.4.1, except that dielectric withstand voltage need not be determined.

4.7.2.6 Temperature cycling. A current measuring system shall be subjected to temperature cycling over the range of temperatures of 3.8.2. The test shall be performed to check the system accuracy in accordance with 4.7.2.2 as follows:

(a) Determine accuracy at reference conditions.
(b) Condition the system at minimum temperature for 8 hours.
(c) Determine accuracy at the minimum temperature.
(c) Condition the system at reference temperature 26.7 ± 2°C for 8 hours.
(e) Repeat (b), (c), and (d) twice for a total of three cycles to minimum temperature and return to reference temperature.
(f) Condition the system at maximum temperature (50°C) for 8 hours.
(g) Determine accuracy at maximum temperature (50°C).
(h) Condition the system at reference temperature 26.7 ± 2°C for 8 hours.

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(i) Repeat (f), (g), and (h) twice for a total of three cycles to maximum temperature and return to reference temperature.
(j) Maximum change in indication between observations at minimum and maximum temperatures shall be considered to be the effect of temperature.

4.7.2.7 Inclination. Using the fixture as specified in 4.7.2.4, the system shall be inclined forward, backward, to the right, and to the left to an angle of 60 degrees from the vertical to test compliance with accuracy requirement of 3.7.1. The accuracy shall be checked at busbar current and ampere-hours reading as follows:

Charge busbar current (amperes)	Discharge busbar current (amperes)	Ampere-hours (minimum)
50	50	5

4.7.3 Insulation resistance. Insulation resistance test shall be conducted in accordance with method 302 of MIL-STD-202. Test conditions shall be as follows:

(a) Test potential: 500 volts plus or minus 10 percent d.c.
(b) Points of measurement: As specified in 3.6.
(c) Electrification time: 60 seconds minimum.

4.7.3.1 Humidity. An ammeter ampere-hour meter and a totalizing ampere-hour indicator shall be stored for 168 hours at 95 percent ± 5 percent relative humidity at 31 ± 2°C. Upon completion of this period, the components shall be tested as specified in 4.7.3 for compliance with 3.8.3. The test shall be performed within 5 minutes after completion of the humidity period. Following the test, the components shall be carefully examined for any signs of material damage and corrosion.

4.7.4 Dielectric withstand voltage. Dielectric withstand voltage shall be conducted in accordance with Method 301 of MIL-STD-202. Test conditions shall be as follows:

(a) Magnitude of test voltages. Magnitude shall be as specified in 3.6.
(b) Nature of test voltage. The test voltage shall approximate a true sine wave of a frequency not less than the rated frequency of the circuit under test.
(c) Points of measurement. Points of measurement shall be as specified in 3.6.
(d) Duration of test. Duration shall be 60 seconds, minimum. The test voltage shall be removed by adjustment of its value to zero, not by sudden interruption.
(e) Definition of failure. Any evidence of arcing, corona (visible or odorous), flashover or punctured insulation shall be interpreted as failure of the test.
(f) Dielectric withstand voltage test. This test shall be conducted after all other tests have been completed.

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4.7.5 Temperature rise (ammeter and ampere-hour meter only). The ammeter and ampere-hour meter shall be operated with meters and indicator or simulated impedances connected and also with a shunt connected. The shunt energy shall be varied to determine the value (within limits corresponding to busbar currents from 0 to 8,000 amperes) resulting in maximum power dissipation in the ammeter and ampere-hour meter circuits. The ambient temperature shall be maintained at 50°C. Measurement of temperatures of circuit parts shall be performed as specified in MIL-E-917. The power consumed by the control unit shall be measured to comply with 3.7.2.

4.7.6 Calibration demonstration. Resistance shall be inserted between the ammeter and the shunt and between the ampere-hour and the totalizing ampere-hour indicators to simulate various cable lengths. The resistance shall be adjusted to 0.2, 0.5, zero, 1 and 3.5 ohms per conductor and compliance with 3.7.5 shall be demonstrated.

4.8 Maintainability demonstration test. The time required to accomplish normal corrective maintenance and preventive maintenance tasks shall be demonstrated. The elapse time and manhours required to replace each of the onboard repair parts shall be reported. The elapsed time and manhours to perform each proposed preventive maintenance task shall also be reported. These time recordings may be obtained where possible from the maintenance actions associated with other tests required by this specification in order to minimize the amount of special effort required to obtain the maintainability data. The time data shall include the fault correction time for corrective maintenance tasks and servicing time for preventive maintenance exclusive of any subsequent test time. A maintainability demonstration report shall be submitted which contains the data recorded, analysis of the data, and the results of any retests if applicable. The procedure for the maintainability demonstration shall be the same as the procedures that will be in the equipment technical manual.

4.9 Reliability. Reliability tests shall be in accordance with MIL-STD-781, test plan III, test level B.

4.10 Enclosures. The enclosures shall be inspected to determine conformance with 3.5.1 and 3.5.2.

4.11 Inspection of packaging. Preservation, packaging, packing, and marking shall be inspected to assure compliance with section 5 of this specification.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)

5.1 Preservation-packaging, packing and marking. Preservation-packaging shall be level A or C, packing level A, B, or C and marking in accordance with MIL-E-17555 for equipments and stock repair parts. Onboard repair parts shall be preserved-packaged level A, packed level C, and marked in accordance with MIL-E-17555.

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6. NOTES

6.1 Intended use. This specification is intended to cover a digital current measuring system used to measure and display the instantaneous value and time integral of d.c. in a storage battery circuit. The ammeter continuously measures and displays the current into or out of the battery. The ampere-hour meter continuously records the ampere-hours entering or leaving the battery; therefore, it can be used to monitor the battery state of charge. The totalizing ampere-hour meter provides a summation of ampere-hours used over the life of the battery.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class and type (see 1.2.1 and 1.2.2).
(c) When cable is required (see 3.4).
(d) If maintainability demonstration test is required (see 4.8).
(e) Level of shipment required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirements title	Applicable DID no.	Option
3.11	Technical manual	DI-M-2044	Type II of MIL-M-15071
3.11.1	Drawings, engineering and associated lists	DI-E-7031	Level 3 Design activity designation - contractor Drawing No. - contractor
4.3.1	First article inspection report	DI-T-4902	---
4.6	Test plan	UDI-T-23732	---
4.6	Test report	DI-T-2072	---

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. *This does not apply to specific data which may be required for* each contract, regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to reply on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of reference documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the system is shipped.

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